The 15 Core Papers for Forecasting Volatility and Tail Risk

Deep Literature Review, Methodological Audit, and Project Mapping

GARCH-EVT | Realized GARCH | Quantile Regression | VaR/ES Evaluation

Status and verification standard
This is a source-grounded working literature review designed for team verification. Where full text or an author preprint was accessible, the discussion uses the paper's own equations, sample design, and stated conclusions. Where only a publisher abstract or partial text was accessible, the review deliberately avoids inventing unverified details and labels the paper accordingly. The final submitted literature review should still be checked paper-by-paper against the PDFs your team retains.

Prepared to support the research paper: Forecasting Volatility and Tail Risk During Extreme Market Events.

Executive overview: how the 15 papers fit together

These papers are not equally important for the same reason. Some define the models we actually estimate; some determine how we are allowed to evaluate them; some place hard constraints on our novelty claim; and some directly motivate robustness checks that emerged from the code and results audit.

Paper	Primary role in our project	Verification basis
McNeil & Frey (2000)	Governing conditional GARCH-EVT framework	Publisher abstract + accessible full-text copy located
Hansen, Huang & Shek (2012)	Governing Realized GARCH framework	Full working-paper text / publisher PDF
Engle & Manganelli (2004)	Direct dynamic quantile / DQ framework	Full UCSD discussion-paper PDF
Zikes & Barunik (2016)	Closest foundation for QR-Full predictor logic	Published abstract + open working-paper route
Watanabe (2012)	Skew-t Realized-GARCH VaR/ES robustness	Publisher full-text page
Engle & Ng (1993)	Asymmetry diagnostics and GJR motivation	Publisher/NBER full-text access
Patton (2011)	Proxy-robust volatility-loss comparison; QLIKE	Publisher full-text excerpts / abstract
Gerlach & Wang (2020)	Closest broad realized-measure VaR/ES comparator	Open arXiv full text
Bee, Dupuis & Trapin (2018)	Realized + quantile + EVT hybrid; novelty constraint	Open-access full-text copy
Trapin (2018)	Extremal dependence and leverage	Accessible preprint / publisher abstract
Paul & Sharma (2017)	Realized-GARCH-EVT VaR comparator	Publisher abstract + methodology/findings
Paul & Sharma (2018)	Realized-GARCH-EVT VaR/ES + realized-measure sensitivity	Publisher + accessible full-text copy
Fissler & Ziegel (2016)	Joint elicitability of VaR and ES	Open arXiv full text
Wu & Cai (2024)	Recent robust realized-measure VaR/ES comparison	Publisher full-text page
Chen, Koike & Shau (2024)	Overnight information in international tail forecasting	Publisher full-text page

The single most important synthesis

The literature does not support a simple league-table view of 'best model'. It supports separating (i) conditional-scale accuracy, (ii) tail-quantile calibration, (iii) proper forecast scoring, and (iv) crisis-state robustness. That separation matches the central pattern emerging from the project's current results.

Reading order

- 1-3 define the three modelling philosophies: McNeil-Frey, Realized GARCH, CAViaR.

- 4-6 justify the actual predictor/asymmetry/density choices in the implemented models.
- 7 and 13 determine how volatility and ES forecasts should be evaluated.
- 8-12 constrain the research gap by showing that hybrids and close comparisons already exist.
- 14-15 provide the most relevant recent evidence on realized-measure choice and international information timing.

1. McNeil, A. J. & Frey, R. (2000), 'Estimation of tail-related risk measures for heteroscedastic financial time series: an extreme value approach', Journal of Empirical Finance 7, 271-300.

Verification status
Publisher abstract and an accessible article copy were reviewed. The core two-stage framework, VaR/ES objective, and empirical conclusion are directly supported. Exact paper-specific sample details should still be checked against the team's PDF before final submission.

Why this paper matters

This is the governing methodological paper for our GARCH-EVT branch. It formalizes the idea that financial tail risk should not be estimated from raw returns as if they were iid. First filter predictable heteroskedasticity; then model only the extreme tail of the standardized innovation distribution.

Research problem and contribution

The problem is that unconditional EVT applied directly to returns confounds two sources of risk: time-varying volatility and the shape of the residual tail. McNeil and Frey propose a conditional approach that uses a GARCH-type model to estimate current conditional scale and EVT to estimate the innovation tail. This yields conditional VaR and conditional Expected Shortfall rather than static unconditional tail measures.

Theoretical and methodological framework

Write returns as r_t = mu_t + sigma_t z_t. The first stage estimates mu_t and sigma_t from a heteroskedastic time-series model. If the filter is adequate, z_t should be much closer to a stable innovation sequence than raw returns.

The second stage deliberately refuses to fit the entire distribution of z_t. Only sufficiently extreme observations are used. Peaks over threshold theory motivates approximating tail exceedances by a Generalized Pareto Distribution (GPD).

For a left-tail equity-risk application, it is cleanest to work on a positive loss scale L_t = -z_t. A high threshold u is chosen and excesses L_t-u conditional on L_t>u are fitted by a GPD with shape xi and scale beta.

Conditional risk forecasts are obtained by combining the forecast conditional mean and scale with the fitted standardized tail quantile. This decomposition is conceptually central because EVT changes the innovation quantile, not the volatility recursion itself.

r_t = mu_t + sigma_t z_t

L_t = -z_t ;   L_t - u | L_t > u ~ GPD(xi, beta)

VaR_(alpha,t+1) = mu_hat_(t+1) + sigma_hat_(t+1) q_hat_(alpha,EVT)

Data and empirical design

The paper applies the method to heteroskedastic financial return series and evaluates conditional quantile and tail-mean forecasts. The published abstract emphasizes both VaR and Expected Shortfall. Because our current web-accessible source did not cleanly expose every table/sample field, those exact empirical details should be verified directly from the paper PDF before being repeated verbatim.

Estimation, forecasting, and evaluation

Pseudo-maximum-likelihood GARCH fitting is used for the conditional mean/scale stage, followed by EVT estimation for the innovation tail. The paper then backtests conditional VaR and ES forecasts against alternative distributional approaches. The central statistical logic is that the first-stage filter and second-stage tail model solve different parts of the problem.

Main findings

- The filtered-EVT approach produces strong conditional tail-risk forecasts and demonstrates that heavy residual tails remain economically important after heteroskedasticity is modelled.
- Expected Shortfall is treated as a genuine tail severity measure rather than merely another quantile.
- The paper establishes the now-standard 'conditional EVT' template used in financial risk management: volatility filter first, EVT second.

Strengths

- The framework has a clear decomposition between conditional volatility and tail shape.
- It uses EVT where EVT is theoretically strongest - in the extreme tail rather than over the whole distribution.
- It provides both VaR and ES, allowing the magnitude of losses beyond VaR to be studied.

Limitations and cautions

- The method is only as trustworthy as the first-stage filter. Residual serial dependence or remaining volatility clustering can contaminate EVT inference.
- Threshold choice creates the standard bias-variance trade-off: too low violates the asymptotic GPD approximation; too high leaves too few exceedances.
- The tail sample is intrinsically data-hungry, which becomes severe in rolling-window implementations.
- Stationarity of the standardized tail over a long expanding history is an assumption that can be strained by structural breaks.

Direct implications for our project

- Our implementation should be described explicitly as a McNeil-Frey-style conditional EVT model with an asymmetric skew-t GJR filter rather than a generic EVT model.
- The causal-residual repair is methodologically important: every historical standardized residual entering the tail fit must itself be computed from parameters available by the forecast origin.
- GARCH-EVT and GJR-ST share the same sigma forecast in our implementation; the comparison therefore isolates the value of replacing the parametric skew-t tail by an EVT tail.
- Residual, squared-residual, and exceedance-indicator diagnostics belong in the final paper because filtered EVT is not automatically valid after division by sigma.

What we should cite this paper for

- Why the project uses a two-stage GARCH + POT/GPD tail model.

- Why VaR and ES can be obtained from a conditional volatility forecast combined with an EVT innovation tail.
- Why tail modelling should be applied to standardized residuals rather than raw heteroskedastic returns.

What we should not overclaim from it

- Do not cite it as proof that EVT must outperform every strong skew-t GARCH benchmark in every market.
- Do not imply that the paper removes the need to diagnose the first-stage filter or choose the threshold carefully.
- Do not present our exact expanding-window/causal-residual implementation as if it were prescribed identically by McNeil and Frey.

Primary source / verification link: https://doi.org/10.1016/S0927-5398(00)00012-8

2. Hansen, P. R., Huang, Z. & Shek, H. H. (2012), 'Realized GARCH: A Joint Model for Returns and Realized Measures of Volatility', Journal of Applied Econometrics 27, 877-906.

Verification status

Full working-paper text / publisher PDF was reviewed. Equations, leverage function, empirical sample, and in-/out-of-sample comparisons are directly supported.

Why this paper matters

This is the governing paper for the project's Realized GARCH implementation. The key contribution is not merely adding lagged realized variance to a GARCH equation; it is building a joint model for returns, latent conditional variance, and an observed realized measure through a measurement equation.

Research problem and contribution

The paper asks whether high-frequency realized measures can be integrated into a parsimonious GARCH framework without abandoning the tractability of classical conditional-variance models. The authors introduce linear and log-linear Realized GARCH classes and show that the measurement equation creates a direct statistical link between the latent variance driving returns and the realized measure computed from intraday data.

Theoretical and methodological framework

The generic Realized GARCH structure contains a return equation, a GARCH equation for latent variance, and a measurement equation for the realized measure. The measurement equation is the defining innovation.

The paper stresses that the realized measure need not be an unbiased estimate of the latent h_t. This is important for our project because session-only realized variance and close-to-close return variance are not identical objects.

In the log-linear model, log h_t responds to lagged log h and lagged log realized measure. The measurement equation links log x_t to log h_t and a leverage function of the standardized return shock.

The baseline leverage function is quadratic: tau(z_t) = tau_1 z_t + tau_2(z_t^2 - 1). This captures dependence between contemporaneous return shocks and the realized measurement innovation and induces leverage-like dynamics.

The authors show that Realized GARCH implies ARMA-type representations for volatility and realized measures, retaining much of the interpretability of standard GARCH while using far more intraday information.

r_t = sqrt(h_t) z_t

log h_t = omega + beta log h_(t-1) + gamma log x_(t-1)

log x_t = xi + phi log h_t + tau_1 z_t + tau_2(z_t^2 - 1) + u_t

Data and empirical design

The empirical sample spans 1 January 2002 to 31 August 2008, with 2002-2007 used in-sample and the first eight months of 2008 used out-of-sample. The study reports results for 29 assets based on DJIA stocks and an exchange-traded index fund. A realized kernel is used as the realized measure to reduce sensitivity to market microstructure noise.

Estimation, forecasting, and evaluation

The paper estimates alternative linear/log-linear specifications and compares models with and without leverage terms. Out-of-sample likelihood comparisons plug in in-sample parameter estimates. Diagnostic comparisons of robust and conventional standard errors and residual scatterplots are used to assess misspecification.

Main findings

- The log-linear formulation fits the assumed likelihood structure substantially better than the linear formulation.
- The quadratic leverage function materially improves the specification and the residual relationship between return shocks and measurement errors.
- Across the empirical assets, Realized GARCH provides substantially better in-sample and out-of-sample partial likelihood than standard GARCH models using daily returns only.
- Parameter estimates are reported as remarkably similar across many assets, suggesting a fairly stable empirical structure.

Strengths

- It provides a coherent joint model rather than an ad hoc GARCH-X regression.
- The measurement equation makes high-frequency information statistically explicit.
- The paper offers both structural insight and practical diagnostic evidence for the log-linear/leverage specification.

Limitations and cautions

- The original empirical comparison is primarily likelihood-based; it is not identical to our VaR/ES forecast contest.
- The paper does not imply that any realized measure automatically improves extreme-tail calibration.
- The quality of the high-frequency measure is an additional source of model risk.
- The original out-of-sample design does not use the same repeated walk-forward parameter refits we require for a strict modern forecasting comparison.

Direct implications for our project

- Our three-equation implementation should be described as a log-linear Realized-GARCH-style model with the quadratic measurement leverage function.
- Our current plan to make Realized GARCH parameters genuinely walk-forward is stricter than the current full-sample-parameter implementation and makes the comparison fairer against GJR.
- The project should separate the benefit of realized information for h_t from the adequacy of the chosen return innovation density for VaR.
- The Nikkei missing-realized-measure episode directly illustrates the extra failure mode created by relying on a measurement stream.

What we should cite this paper for

- Why Realized GARCH is fundamentally different from inserting RV as an explanatory variable in standard GARCH.
- The three-equation model and the quadratic leverage function.
- Why high-frequency realized measures can improve information about latent conditional variance.

What we should not overclaim from it

- Do not cite the paper as evidence that Realized GARCH must beat GARCH on 1% VaR.
- Do not describe our Hansen-Lunde scaling, missing-data recursion, or Student-t density choices as if they were all dictated by the original paper.
- Do not infer that a better Realized-GARCH likelihood automatically means better tail-risk calibration.

Primary source / verification link: https://doi.org/10.1002/jae.1234

3. Engle, R. F. & Manganelli, S. (2004), 'CAViaR: Conditional Autoregressive Value at Risk by Regression Quantiles', Journal of Business & Economic Statistics 22, 367-381.

Verification status

Full UC San Diego discussion-paper PDF was reviewed. Model classes, optimization approach, DQ test logic, empirical assets, and key out-of-sample findings are directly supported.

Why this paper matters

This paper establishes the financial-risk logic behind direct conditional-quantile modelling. It moves the target from an entire return density to the conditional quantile itself, which is exactly why quantile regression is a genuinely different modelling philosophy in our project.

Research problem and contribution

The authors argue that most VaR methods first estimate a return distribution and then infer a quantile indirectly. CAViaR instead treats VaR as a dynamic conditional quantile process. This relaxes the need to specify a complete innovation density and permits asymmetry directly in the quantile dynamics.

Theoretical and methodological framework

The paper specifies several dynamic quantile forms, including Adaptive, Symmetric Absolute Value, Asymmetric Absolute Value, Asymmetric Slope, and an Indirect GARCH form. The Asymmetric Slope model is especially important because positive and negative lagged returns can have different effects on the conditional quantile.

Parameters are estimated by minimizing the nonlinear regression-quantile objective. Because the objective is non-differentiable, the paper uses a differential evolutionary genetic algorithm for numerical optimization.

A central contribution is the Dynamic Quantile (DQ) test. Correct VaR calibration requires more than the right unconditional hit rate: the hit sequence should be unpredictable from available information.

The paper's test philosophy maps closely to our evaluation framework. Kupiec-style coverage alone can look acceptable while violations remain dynamically clustered.

$$Q_{\tau}(r_t \mid F_{(t-1)}) \text{ is modelled directly rather than recovered from a full conditional density}$$

$$Hit_t = I(r_t < VaR_t) - \alpha$$

$$\text{Correct dynamic calibration requires } E[Hit_t \mid F_{(t-1)}] = 0$$

Data and empirical design

The empirical application uses GM, IBM, and the S&P 500. The study examines several quantile levels, including very low tail probabilities, and contains a 500-trading-day out-of-sample period. The estimation sample includes the October 1987 crash, while the out-of-sample period includes the high-volatility summer of 1998.

Estimation, forecasting, and evaluation

The competing CAViaR specifications are assessed using hit rates, the regression-quantile objective, and multiple forms of the DQ test. Model selection is discussed in terms of rejecting dynamically inadequate specifications and then comparing out-of-sample quantile loss.

Main findings

- Several flexible CAViaR specifications perform well both in and out of sample.
- The Adaptive model can have acceptable unconditional hit counts while failing DQ because hits remain autocorrelated - a direct demonstration of why conditional calibration matters.
- The Asymmetric Slope specification performs strongly for GM and the S&P 500, supporting asymmetric lower-tail dynamics.
- The authors emphasize that different quantile levels may require different models.

Strengths

- It directly targets the object risk managers care about: the conditional quantile.
- The DQ test provides a much richer adequacy check than unconditional coverage.
- The model is semiparametric in the sense that it does not require a full return distribution.

Limitations and cautions

- Standard CAViaR does not by itself produce ES.
- Separate quantile regressions can lead to crossing when several quantile levels are estimated independently.
- The original CAViaR models use relatively parsimonious dynamic regressors; our QR-Full uses a richer external-information set and is not simply a CAViaR replication.
- Very deep quantiles remain statistically difficult because effective tail information is scarce.

Direct implications for our project

- Our QR branch should be motivated as direct conditional-quantile estimation, not as an alternative volatility equation.
- The DQ test belongs in the final VaR evaluation because a correct breach count is not enough.
- Our finding that abrupt crises create clusters of misses is exactly the type of failure the DQ perspective is designed to detect.
- QR's reconstructed SigmaHat field must remain excluded from volatility losses.

What we should cite this paper for

- Why VaR can be modelled directly as a conditional quantile.
- Why DQ should supplement unconditional and first-order conditional coverage tests.
- Why asymmetric quantile dynamics are economically plausible.

What we should not overclaim from it

- Do not cite CAViaR as evidence that vanilla QR produces Expected Shortfall.
- Do not claim that the project's predictor-rich QR specification is the same model as Engle-Manganelli CAViaR.
- Do not treat a correct unconditional hit rate as sufficient just because a quantile method is used.

Primary source / verification link: https://doi.org/10.1198/073500104000000370

4. Zikes, F. & Barunik, J. (2016), 'Semi-parametric Conditional Quantile Models for Financial Returns and Realized Volatility', Journal of Financial Econometrics 14, 185-226.

Verification status

Published abstract and open working-paper route were reviewed. The broad model classes, variables, assets, and principal findings are directly supported. Exact coefficient-level claims should be re-checked against the full PDF before submission.

Why this paper matters

This is the closest conceptual foundation for our QR-Full predictor architecture. It shows that realized variance, upside/downside semivariance, jump variation, and option-implied volatility can be used to model conditional return quantiles directly.

Research problem and contribution

The paper asks how conditional quantiles of future returns and future realized volatility vary with ex-post realized measures and option-implied volatility. Rather than estimating a latent volatility process and imposing a return distribution, the authors work in a flexible quantile-regression framework.

Theoretical and methodological framework

For return quantiles, simple linear conditional-quantile regressions use realized and implied information as explanatory variables.

For realized-volatility quantiles, the authors use heterogeneous quantile autoregressions, conceptually related to HAR-style multi-horizon dynamics.

An important contribution is the economically structured predictor set: integrated variance, positive and negative semivariance, jump variation, and option-implied volatility are not arbitrary transformations but distinct measures of market variation and forward-looking risk.

The paper's logic supports blockwise interpretation of predictors rather than treating a large QR model as an unrestricted machine-learning exercise.

Q_tau(r_(t+1) | X_t) = X_t' beta_tau

X_t may include realized variance, downside/upside semivariance, jump variation, and option-implied volatility

Data and empirical design

The empirical analysis uses S&P 500 and WTI Crude Oil futures. These assets allow the authors to test whether the same semiparametric quantile logic works across different markets and whether realized/implied information helps characterize both return tails and realized-volatility distributions.

Estimation, forecasting, and evaluation

The models are estimated using quantile-regression methods and evaluated against established benchmarks. The abstract reports strong performance of simple linear return-quantile models and heterogeneous quantile-autoregressive realized-volatility models, both absolutely and relative to benchmark approaches.

Main findings

- Realized and implied measures contain useful information about conditional financial quantiles.
- Simple linear quantile regressions can perform very well despite avoiding a complete parametric density.
- Signed realized variation and jump information can matter separately from total realized variance.

Strengths

- It gives our QR variable blocks a clear economic and econometric precedent.
- It connects realized-volatility research to direct tail-quantile forecasting.
- It uses both return and volatility quantiles, clarifying that the same quantile-regression machinery can target different objects.

Limitations and cautions

- The paper is based on two main futures markets; our six international equity-index setting is broader.
- Predictor usefulness is not guaranteed to be stable across markets or sample periods.
- Using many highly correlated realized predictors can create instability, so collinearity control remains important.
- The published paper does not validate every macro variable in our QR-Full specification.

Direct implications for our project

- Use it to justify LogRV, semivariance/asymmetry, jump, and implied-volatility predictor blocks.
- Present our predictor selection as literature-driven rather than data-mined.
- Use it to explain why QR can use richer market information than a low-dimensional latent-volatility recursion.

What we should cite this paper for

- The relevance of realized variance, semivariance, jump variation, and implied volatility for conditional quantiles.
- The legitimacy of simple linear quantile regressions as risk-management tools.
- The rationale for predictor blocks in QR-Full.

What we should not overclaim from it

- Do not cite it as direct support for our exact 13-predictor specification or macro variables that the paper does not study.
- Do not imply its results guarantee strong 1% performance in every market.
- Do not treat realized-volatility quantile models in the paper as equivalent to our return-VaR regressions.

Primary source / verification link: https://doi.org/10.1093/jjfinec/nbu029

5. Watanabe, T. (2012), 'Quantile Forecasts of Financial Returns Using Realized GARCH Models', Japanese Economic Review 63, 68-80.

Verification status

Publisher full-text page was reviewed. Distributional alternatives, main S&P 500 findings, and realized-kernel conclusion are directly supported.

Why this paper matters

This paper became more important after our code/results audit than it appeared initially. It directly asks how Realized GARCH performs for VaR and ES when the conditional return density is Normal, Student-t, or skewed Student-t.

Research problem and contribution

The research problem is exactly the confounding issue in our current Realized-GARCH result: does a realized-information volatility model fail in the tail because the volatility state is poor, or because the innovation distribution is too restrictive?

Theoretical and methodological framework

The paper starts from Realized GARCH and changes the conditional return distribution while keeping the realized-information logic intact.

A symmetric Student-t allows heavy tails but not left-right asymmetry. A skewed Student-t allows both heavy tails and asymmetric tail probability.

This comparison is especially valuable because a model can estimate h_t well yet produce VaR that is systematically too narrow if the standardized return density is misspecified.

r_t = mu_t + sqrt(h_t) z_t, with z_t alternatively Normal, Student-t, or skewed Student-t

VaR and ES depend jointly on h_t and the chosen innovation quantile/tail mean

Data and empirical design

The main empirical application is the S&P 500 with high-frequency realized measures. The paper also compares plain realized volatility with a realized kernel designed to address market microstructure noise.

Estimation, forecasting, and evaluation

The paper estimates Realized GARCH under alternative innovation distributions and evaluates one-day-ahead VaR and Expected Shortfall performance. The key design holds the broad realized-information framework fixed while changing the return-density assumption.

Main findings

- Realized GARCH with skewed Student-t innovations performs better than Realized GARCH with Normal or symmetric Student-t innovations in the S&P 500 application.

- The skew-t Realized GARCH also outperforms an EGARCH benchmark using daily returns only.
- Using a realized kernel instead of plain RV does not improve performance in this application.

Strengths

- It isolates the role of the conditional return density within Realized GARCH.
- It directly studies VaR and ES rather than only volatility likelihood.
- It provides a clean literature precedent for exactly the robustness model we need.

Limitations and cautions

- The main result is from a single primary equity index, so it should not be generalized mechanically to all six of our markets.
- The paper's realized-measure conclusion does not prove that realized kernels are useless in general.
- Different skew-t parameterizations and estimation details can matter.

Direct implications for our project

- We should add a Realized-GARCH-skew-t robustness model before interpreting poor Realized-GARCH VaR as a failure of realized information.
- If skew-t materially improves coverage, the paper's main story becomes 'variance information was good; the symmetric density was not'.
- If skew-t does not help, the separation between volatility accuracy and tail calibration becomes even stronger.

What we should cite this paper for

- Why a symmetric Student-t Realized GARCH may underperform in the lower tail.
- Why a skew-t Realized GARCH is a necessary robustness model.
- Why realized-measure quality and return-density quality are separate modelling choices.

What we should not overclaim from it

- Do not cite Watanabe as proof that skew-t will necessarily fix our Realized-GARCH VaR.
- Do not claim realized kernels never help based on one application.
- Do not replace the pre-specified Student-t model silently; use skew-t as an explicit robustness extension.

Primary source / verification link: https://doi.org/10.1111/j.1468-5876.2011.00548.x

6. Engle, R. F. & Ng, V. K. (1993), 'Measuring and Testing the Impact of News on Volatility', Journal of Finance 48, 1749-1778.

Verification status
Publisher/NBER text was reviewed. News-impact curve, asymmetry diagnostics, Japanese-return application, and comparative conclusion are directly supported.

Why this paper matters

This paper is the load-bearing justification for using an asymmetric volatility filter before tail modelling. It gives formal diagnostic tools for detecting whether positive and negative return news has different volatility consequences.

Research problem and contribution

The authors introduce the news-impact curve as a way of visualizing how new shocks map into next-period volatility and develop diagnostic tests emphasizing asymmetry. They compare multiple ARCH-type models using Japanese stock returns.

Theoretical and methodological framework

The sign-bias test asks whether the sign of an innovation contains variance information omitted by the fitted model.

Negative-size and positive-size bias tests ask whether the magnitude of negative or positive innovations has residual explanatory power.

The news-impact curve provides a common way to compare the implied volatility response across otherwise different ARCH specifications.

The paper also uses a partially nonparametric ARCH benchmark, which is important because it does not assume one parametric asymmetry shape must be correct.

GJR-style variance effect: alpha epsilon^2 for positive shocks versus (alpha + gamma) epsilon^2 for negative shocks

News-impact diagnostics test whether the fitted variance model has absorbed sign and size asymmetry

Data and empirical design

The empirical application uses daily Japanese stock returns. The NBER summary and published abstract report similar qualitative results on a pre-crash subsample, though weaker than in the full sample.

Estimation, forecasting, and evaluation

The authors estimate several ARCH-family models and compare their news-impact curves and residual asymmetry diagnostics.

Main findings

- Negative shocks have a stronger volatility impact than positive shocks of comparable magnitude.
- The GJR model is judged the best parametric model in their comparison.
- EGARCH captures much of the asymmetry, but the paper reports evidence that EGARCH can imply excessively variable conditional variance.

Strengths

- It links model choice to diagnostic evidence rather than information criteria alone.
- It provides both graphical and formal asymmetry diagnostics.
- It directly supports the economic importance of leverage-type behavior.

Limitations and cautions

- The main empirical application is one market and period.
- These are volatility-asymmetry diagnostics, not tail-calibration tests.
- Passing Engle-Ng diagnostics does not prove extreme residuals are iid or that the innovation density is correctly specified.

Direct implications for our project

- Our use of GJR should be motivated by observed asymmetry rather than just convention.
- Engle-Ng diagnostics should be reported as evidence for asymmetric scale dynamics.
- The paper helps explain why the GJR-skew-t benchmark is intentionally strong before we ask EVT to add anything.

What we should cite this paper for

- Why negative and positive return shocks should be allowed to affect volatility differently.
- Why the GJR model is a natural primary asymmetric specification.
- Why news-impact curves/sign-bias diagnostics belong in the model-validation stage.

What we should not overclaim from it

- Do not cite it as evidence that GJR is universally superior in all markets or periods.
- Do not treat Engle-Ng tests as EVT tail-independence tests.
- Do not infer that a volatility-asymmetry result automatically implies asymmetric standardized residual tails.

Primary source / verification link: https://doi.org/10.1111/j.1540-6261.1993.tb05127.x

7. Patton, A. J. (2011), 'Volatility Forecast Comparison Using Imperfect Volatility Proxies', Journal of Econometrics 160, 246-256.

Verification status

Publisher abstract and detailed full-text excerpts were reviewed. The robust-loss result, role of MSE/QLIKE, and IBM empirical illustration are directly supported.

Why this paper matters

This paper governs how we should evaluate the volatility lane. Conditional variance is latent even ex post, so all volatility comparisons rely on imperfect proxies. Patton shows that the choice of loss function determines whether proxy noise can distort model rankings.

Research problem and contribution

The paper asks which loss functions preserve the ranking of competing variance forecasts when the observed volatility proxy is conditionally unbiased but noisy. It derives necessary and sufficient conditions for robustness and studies widely used proxies such as squared returns, intraday range, and realized variance.

Theoretical and methodological framework

A volatility forecast may be correct yet look inferior if the evaluation loss interacts badly with proxy noise.

Patton derives a class of losses whose ranking is robust under the stated proxy conditions. Two important special cases are MSE and QLIKE.

QLIKE is especially attractive because it is scale-sensitive in a way suited to variance forecasts and penalizes severe underprediction strongly while remaining within the robust class.

The paper therefore separates two issues that are often confused: how good the volatility proxy is, and how good the scoring rule is when that proxy is noisy.

$$\text{QLIKE}(v, h) = v/h - \log(v/h) - 1$$

Forecast ranking should be robust to conditionally unbiased proxy noise under the chosen loss

Data and empirical design

The empirical illustration uses IBM return volatility from 1993 to 2003 and compares simple volatility forecasts under alternative proxy/loss combinations.

Estimation, forecasting, and evaluation

The theoretical results are accompanied by empirical comparison methods related to Diebold-Mariano/West-style predictive-accuracy tests.

Main findings

- Some commonly used losses can select an inferior forecast because of proxy noise.

- MSE and QLIKE are important robust special cases under the paper's conditions.
- The empirical application illustrates that proxy/loss choice is not an innocuous reporting decision.

Strengths

- It gives a principled reason for choosing QLIKE rather than selecting a metric post hoc.
- It is directly relevant to our use of a realized-volatility proxy rather than latent true variance.
- It separates forecast-model quality from evaluation-proxy quality.

Limitations and cautions

- Robustness is conditional on assumptions about the proxy, including conditional unbiasedness-type conditions.
- QLIKE does not make a poor realized measure magically correct.
- The paper is about conditional variance forecasts, not VaR or ES.

Direct implications for our project

- QLIKE should be the primary volatility loss in our paper, with RMSE/MAE secondary.
- QR must be excluded because its reconstructed SigmaHat is not a genuine variance forecast.
- Realized GARCH's QLIKE advantage should be interpreted only after the realized proxy and out-of-sample parameter design are made fair.

What we should cite this paper for

- Why realized volatility can be used as an imperfect proxy for variance forecast comparison.
- Why QLIKE is an appropriate primary volatility loss.
- Why volatility ranking can depend on the evaluation loss.

What we should not overclaim from it

- Do not cite Patton as support for using QLIKE on a quantity that is not a variance forecast.
- Do not claim QLIKE eliminates all measurement-error concerns.
- Do not use it to justify VaR or ES scoring rules.

Primary source / verification link: https://doi.org/10.1016/j.jeconom.2010.03.034

8. Gerlach, R. & Wang, C. (2020), 'Semi-parametric Dynamic Asymmetric Laplace Models for Tail Risk Forecasting, Incorporating Realized Measures', International Journal of Forecasting 36, 489-506.

Verification status
Open arXiv full text was reviewed. Model equations, estimation strategy, simulation design, forecasting window, competitor set, and main conclusions are directly supported.

Why this paper matters

This is one of the closest existing papers to our broad conceptual comparison. It combines direct semiparametric VaR/ES dynamics with realized measures and explicitly benchmarks against GARCH, Realized GARCH, EVT, and related models.

Research problem and contribution

The authors extend Taylor's joint VaR-ES semiparametric framework so that realized measures drive the dynamics of tail risk. Their central idea is that realized measures may be a more informative driver of the tail state than lagged daily absolute returns.

Theoretical and methodological framework

The proposed ES-CAViaR-X class replaces or augments lagged daily-return information with a realized measure X_(t-1) in the VaR equation.

VaR and ES are modelled jointly using an asymmetric-Laplace-based likelihood whose negative log-likelihood corresponds to a strictly consistent joint VaR-ES scoring structure.

The paper proposes both autoregressive and exponential relationships between VaR and ES, with constraints designed to prevent ES from crossing VaR.

Both maximum likelihood and adaptive Bayesian MCMC are considered. A simulation study generally favors the Bayesian approach in bias/precision.

Q_t = beta_0 + beta_1 X_(t-1) + beta_2 Q_(t-1)

VaR and ES are estimated jointly rather than deriving ES from a separate ad hoc rule

Data and empirical design

The paper studies seven financial market indices in the published version; the open preprint also discusses two individual assets. The out-of-sample period is 2008-2016 and therefore includes the global financial crisis. Multiple realized measures, including realized variance and realized range variants, are considered.

Estimation, forecasting, and evaluation

The proposed models are estimated by MLE and adaptive Bayesian MCMC, with the Bayesian approach used in the main forecasting exercise after simulation evidence. Forecasts are one-day ahead and compared with a wide range of parametric, nonparametric, and semiparametric competitors.

Main findings

- Realized-measure-driven ES-CAViaR-X models perform favorably relative to traditional competitors.
- Subsampled realized variance and subsampled realized range are especially strong realized inputs in their study.
- The paper illustrates that realized information can improve not only volatility models but direct tail-risk dynamics.

Strengths

- It is a genuine broad comparison across GARCH, Realized GARCH, EVT, and semiparametric VaR/ES models.
- It uses modern joint VaR-ES scoring logic.
- It studies a long OOS period including the GFC.

Limitations and cautions

- Its model is much richer than our vanilla separate-quantile QR, so it is not a one-for-one benchmark.
- The empirical design and realized measures differ from ours.
- Strong realized-measure performance in their framework does not imply Realized GARCH itself must be the best model.

Direct implications for our project

- This paper sharply limits novelty claims: realized-measure semiparametric tail models have already been compared with GARCH, Realized GARCH, and EVT.
- Our contribution should therefore be framed as a controlled six-market comparison of information philosophies, not as the first comparison of these broad families.
- It also supports the idea that QR could be extended to a genuine joint VaR-ES framework in future work.

What we should cite this paper for

- Closest broad comparator for our three-family conceptual design.
- Why realized measures can drive tail risk directly, not only latent volatility.
- Why modern tail-risk evaluation should consider VaR and ES jointly.

What we should not overclaim from it

- Do not claim our project is the first to compare GARCH, Realized GARCH, EVT, and semiparametric tail methods.
- Do not present their realized-measure result as evidence that our specific QR-Full must win.
- Do not conflate ES-CAViaR-X with ordinary linear quantile regression.

Primary source / verification link: https://doi.org/10.1016/j.ijforecast.2019.07.003

9. Bee, M., Dupuis, D. J. & Trapin, L. (2018), 'Realized Extreme Quantile: A Joint Model for Conditional Quantiles and Measures of Volatility with EVT Refinements', Journal of Applied Econometrics 33, 398-415.

Verification status

Open-access full-text copy was reviewed. Model structure, measurement equation, simulation/QML design, 17-index empirical analysis, rolling OOS design, and main conclusions are directly supported.

Why this paper matters

This paper is a crucial novelty constraint. It already combines realized measures, conditional quantiles, a measurement equation, and EVT refinements within one coherent model.

Research problem and contribution

The authors ask whether high-frequency realized measures can improve estimation and forecasting of extreme conditional quantiles directly, rather than first estimating a latent variance as in Realized GARCH.

Theoretical and methodological framework

The Realized Extreme Quantile (REQ) framework contains a return equation, a conditional-quantile equation, and a measurement equation linking the observed realized measure to the latent conditional quantile.

This is conceptually analogous to Realized GARCH's measurement equation, but the latent state is the quantile rather than conditional variance.

The authors use quasi-maximum likelihood and validate the estimator in simulation.

The EVT refinement is used to extrapolate more extreme quantiles from the fitted quantile structure, creating a hybrid of QR and extreme-value methods.

Observed realized measure x_t is linked through a measurement equation to a latent conditional quantile state

REQ combines quantile dynamics + measurement equation + EVT refinement

Data and empirical design

The empirical study covers 17 indices. The paper reports an in-sample period of 2000-2004 and out-of-sample analyses over 2005-2009 and 2010-2014 in parts of the study. For the rolling forecast comparison, a window of 2,000 observations with daily updating is used.

Estimation, forecasting, and evaluation

REQ specifications using low-frequency and high-frequency realized measures are compared with corresponding quantile models without EVT and with lower-frequency information. VaR forecasts are evaluated at extreme levels, including 1%.

Main findings

- High-frequency realized measures contain information about daily extreme quantiles beyond lower-frequency measures.
- The measurement-equation structure can add value when a valid realized proxy is supplied.
- Out-of-sample results generally favor high-frequency realized information and show incremental value from the hybrid structure in many comparisons.

Strengths

- It demonstrates a genuine synthesis of realized measurement, quantiles, and EVT.
- It uses a broad cross-sectional index set.
- It explicitly tests both the value of EVT and the value of high-frequency information.

Limitations and cautions

- The hybrid is more complex than our intentionally separate-model comparison.
- Its rolling-window and realized-measure design differ from ours.
- Because the model itself combines the ingredients, it answers a different question from our 'which information philosophy wins under a common protocol?' question.

Direct implications for our project

- Our research gap must never claim that realized measures + QR + EVT have not previously been combined.
- REQ is a natural benchmark/future extension if we later want a single unified model.
- Our value lies in keeping the model philosophies separate enough to identify which information channel drives forecast performance.

What we should cite this paper for

- Why the literature already contains realized-quantile-EVT hybrids.
- Why measurement equations can be used for latent quantiles, not only latent variance.
- Why high-frequency information can matter directly for extreme conditional quantiles.

What we should not overclaim from it

- Do not claim methodological novelty from combining QR, EVT, and realized measures.
- Do not treat REQ as identical to our QR-Full or GARCH-EVT models.
- Do not infer that a unified hybrid is automatically preferable to a transparent comparative design.

Primary source / verification link: https://doi.org/10.1002/jae.2615

10. Trapin, L. (2018), 'Can Volatility Models Explain Extreme Events?', Journal of Financial Econometrics 16, 297-315.

Verification status
Accessible preprint and publisher abstract were reviewed. The paper's extremal-dependence concept, negative-versus-positive extreme asymmetry, high-frequency volatility-process analysis, and main empirical conclusion are directly supported.

Why this paper matters

This paper is especially important for our final EVT diagnostics. Standard residual diagnostics can look satisfactory while the extreme observations themselves still exhibit serial dependence.

Research problem and contribution

The paper reframes volatility-model adequacy through 'extremal dependence': serial dependence specifically among extreme returns. It asks which volatility processes can generate the dependence structure observed in the far tails.

Theoretical and methodological framework

The paper first documents that extreme financial returns are not necessarily serially independent even when ordinary return autocorrelation is weak.

It distinguishes ordinary volatility persistence from extremal dependence. A model can reproduce the second moment reasonably well while missing dependence among tail events.

Negative extreme returns are found to be much more correlated than positive extremes, linking extremal dependence to leverage/asymmetry.

The paper studies theoretical extremal properties of several high-frequency-based volatility processes and then evaluates models empirically.

$$\text{Extremal dependence} = \text{serial dependence in extreme return events, not simply autocorrelation of squared returns}$$

$$\text{Leverage} + \text{extremal dependence are both important for reproducing extreme events}$$

Data and empirical design

The paper contains a large empirical financial analysis alongside theoretical investigation of high-frequency-based volatility processes. Exact asset-by-asset sample details should be verified from the full PDF if they are needed in our final literature table.

Estimation, forecasting, and evaluation

The empirical analysis compares whether volatility models can reproduce the observed extremal dependence structure, not merely whether they fit average volatility well.

Main findings

- Extreme returns show strong and persistent dependence.
- Extreme negative returns are more strongly dependent than extreme positive returns.

- Only models that can generate extremal dependence and include a leverage component appropriately explain extreme events in the paper's empirical analysis.

Strengths

- It adds a tail-specific diagnostic dimension missing from ordinary GARCH residual checks.
- It directly connects leverage and extreme-event behaviour.
- It supports examining crisis clustering rather than relying only on marginal tail fits.

Limitations and cautions

- The paper does not say every POT model must be declustered; the relevance depends on the residual process after filtering.
- It evaluates volatility models through extremal dependence, which is related to but not identical to VaR backtesting.
- Exact empirical implications can differ by model and market.

Direct implications for our project

- Add an exceedance-indicator dependence test to the EVT validation stage.
- Do not assume that standardized residuals are tail-independent merely because Ljung-Box tests on z_t and z_t^2 are clean.
- Interpret any HSI-style clustering as a substantive tail-dynamics issue, not a minor diagnostic nuisance.

What we should cite this paper for

- Why extreme-event dependence deserves separate testing.
- Why leverage remains important even after high-frequency information is introduced.
- Why good volatility fit need not imply adequate extreme-event dynamics.

What we should not overclaim from it

- Do not cite Trapin as proof that our GARCH-EVT residuals are dependent; test them.
- Do not substitute extremal-dependence diagnostics for VaR calibration tests.
- Do not imply the paper endorses one universal volatility specification.

Primary source / verification link: https://doi.org/10.1093/jjfinec/nbx031

11. Paul, S. & Sharma, P. (2017), 'Improved VaR forecasts using extreme value theory with the Realized GARCH model', Studies in Economics and Finance 34, 238-259.

Verification status

Publisher purpose/method/findings sections were reviewed. Sample, six-model comparison, and reported ranking are directly supported.

Why this paper matters

This is the most direct paper showing that Realized GARCH can be used as the volatility filter inside the McNeil-Frey conditional EVT architecture.

Research problem and contribution

The authors ask whether replacing standard GARCH/EGARCH with Realized GARCH in the first stage of conditional EVT improves one-step-ahead VaR forecasts for international stock indices.

Theoretical and methodological framework

The study compares standalone GARCH, EGARCH, and Realized GARCH with GARCH-EVT, EGARCH-EVT, and Realized-GARCH-EVT.

All EVT variants use the McNeil-Frey-style two-stage approach: conditional volatility filtering followed by tail estimation on standardized innovations.

The design isolates whether the choice of first-stage volatility model changes tail-risk performance.

$$\text{Realized-GARCH-EVT} = \text{Realized-GARCH scale filter} + \text{EVT innovation tail}$$

Data and empirical design

The dataset contains daily returns and realized volatilities for 13 international stock indices from 1 January 2003 to 8 October 2014. Forecasts are one-step ahead.

Estimation, forecasting, and evaluation

The authors evaluate six model classes using multiple VaR backtesting/statistical procedures.

Main findings

- Conditional EVT versions outperform the corresponding standalone GARCH-family models in the reported comparisons.
- Standalone Realized GARCH does not outperform GARCH and EGARCH in VaR forecasting.
- Using Realized GARCH inside the EVT framework materially improves VaR performance, with Realized-GARCH-EVT reported as the best overall model among the six.

Strengths

- The cross-sectional sample is much broader than many earlier high-frequency VaR studies.
- It directly separates the realized-volatility filter from the EVT tail layer.

- It is extremely close to a natural extension of our current model set.

Limitations and cautions

- The sample ends in 2014 and therefore does not include COVID or later stress episodes.
- The study focuses on one-step VaR rather than our broader split between volatility accuracy and tail calibration.
- Its model specification, data sources, realized-measure construction, and evaluation protocol differ from ours.

Direct implications for our project

- We cannot claim that combining Realized GARCH and EVT is novel.
- This paper motivates Realized-GARCH-EVT as a future extension rather than the core contribution.
- Its finding also reinforces our interpretation that a strong volatility state can become especially useful when paired with an explicitly flexible tail model.

What we should cite this paper for

- Prior evidence on Realized-GARCH-EVT for international-index VaR.
- Why realized information and EVT can be complementary rather than competing ingredients.
- Why standalone Realized GARCH tail performance should not be equated with the performance of a realized-information filter plus EVT.

What we should not overclaim from it

- Do not claim the paper proves Realized-GARCH-EVT will dominate in our post-2013 sample.
- Do not describe our project as the first cross-market Realized GARCH versus EVT comparison.
- Do not import the paper's reported ranking without acknowledging the different sample and benchmark strength.

Primary source / verification link: https://doi.org/10.1108/SEF-05-2015-0139

12. Paul, S. & Sharma, P. (2018), 'Quantile forecasts using the Realized GARCH-EVT approach', Studies in Economics and Finance 35, 481-504.

Verification status
Publisher page plus accessible article text were reviewed. Five-index design, VaR/ES focus, five realized measures, and reported comparative results are directly supported.

Why this paper matters

This paper extends the Realized-GARCH-EVT comparison beyond VaR to Expected Shortfall and explicitly asks whether the particular realized-volatility estimator changes forecasting performance.

Research problem and contribution

The authors implement two-stage GARCH-EVT, EGARCH-EVT, and Realized-GARCH-EVT models and estimate the realized version using several alternative realized-volatility measures.

Theoretical and methodological framework

The core architecture remains conditional EVT: a volatility filter produces standardized residuals and EVT models the tail.

The realized-information contribution is therefore in the first-stage scale dynamics, while EVT handles the extreme tail.

The paper's multiple-realized-measure design is valuable because it distinguishes 'using intraday information' from 'which intraday estimator is best'.

$$\text{Risk forecast} = \text{conditional scale model} + \text{EVT residual-tail model}$$

$$\text{Realized-measure sensitivity is tested within the same Realized-GARCH-EVT architecture}$$

Data and empirical design

The empirical study generates one-step-ahead VaR and ES forecasts for five European stock indices. The paper describes the high-frequency sample as roughly 11 years and compares five alternative realized measures inside the Realized-GARCH-EVT class.

Estimation, forecasting, and evaluation

Forecasts from Realized-GARCH-EVT are compared with GARCH-EVT and EGARCH-EVT using VaR and ES backtesting/evaluation procedures.

Main findings

- Realized-GARCH-EVT models generally outperform standard GARCH-EVT and EGARCH-EVT in out-of-sample comparisons.
- The choice among the five realized estimators does not materially alter the forecasting ability of the Realized-GARCH-EVT model in their study.

- For ES, the accessible article text reports that Realized-GARCH-EVT variants provide the best forecasts in the great majority of the reported model-index-quantile comparisons.

Strengths

- It examines both VaR and ES.
- It directly studies realized-measure sensitivity.
- It provides another strong constraint on novelty claims.

Limitations and cautions

- The conclusion that realized-estimator choice does not matter is application-specific and should not be generalized universally.
- The paper's ES evaluation methods predate some modern joint VaR-ES scoring practice.
- The study does not include a direct QR family.

Direct implications for our project

- Our frequency and realized-measure robustness section should acknowledge that some prior work finds the realized estimator less important once Realized GARCH-EVT is specified.
- Our results can differ because our primary realized model is not combined with EVT.
- The paper makes Realized-GARCH-EVT an obvious future extension if scope allows.

What we should cite this paper for

- Prior evidence on Realized-GARCH-EVT for both VaR and ES.
- Why realized-measure choice can be tested as a distinct robustness dimension.
- Why realized information may be most effective when combined with a flexible tail layer.

What we should not overclaim from it

- Do not cite it as evidence that five-minute RV is always equivalent to every robust realized estimator.
- Do not treat its ES ranking method as identical to modern Fissler-Ziegel scoring.
- Do not claim its European findings automatically generalize to our US/Europe/Asia panel.

Primary source / verification link: https://doi.org/10.1108/SEF-09-2016-0236

13. Fissler, T. & Ziegel, J. F. (2016), 'Higher Order Elicitability and Osband's Principle', Annals of Statistics 44, 1680-1707.

Verification status

Open arXiv full text was reviewed. The general elicitability framework and the applied result that (VaR, ES) is jointly elicitable are directly supported.

Why this paper matters

This paper determines what constitutes a principled forecast comparison for Expected Shortfall. It is theoretical rather than a finance forecasting application, but its implication for our evaluation framework is fundamental.

Research problem and contribution

A functional is elicitable if there exists a scoring function uniquely minimized in expectation by the true value of that functional. ES alone is not elicitable in the same simple one-dimensional sense, which historically complicated forecast ranking.

Theoretical and methodological framework

The paper generalizes elicitability to multidimensional functionals and provides conditions for strictly consistent scoring functions.

A key result is that some non-elicitable one-dimensional quantities become elicitable when forecast jointly with appropriate companion functionals.

For financial risk, the applied result is that the pair (VaR_alpha, ES_alpha) is jointly elicitable under mild conditions commonly satisfied in risk-management applications.

This creates a rigorous foundation for joint VaR-ES scoring and for semiparametric models that estimate the pair together.

(VaR_alpha, ES_alpha) is jointly elicitable under mild conditions

Proper joint scoring can rank forecasts of VaR and ES together

Data and empirical design

The paper is theoretical and does not contain a conventional market-data forecasting application.

Estimation, forecasting, and evaluation

The methodology is mathematical/statistical decision theory. It characterizes consistent scores through higher-order elicitability and Osband-type arguments.

Main findings

- Joint VaR-ES scoring is theoretically legitimate even though ES alone lacks a simple strictly consistent univariate loss.
- The result provides the foundation for later FZ/FZ0 scoring and joint dynamic VaR-ES models.

Strengths

- It gives a rigorous answer to a subtle but important forecast-evaluation problem.
- It is model-agnostic: the result applies regardless of whether forecasts come from GARCH, EVT, or semiparametric models.

Limitations and cautions

- The paper does not specify a unique empirical score or a unique VaR-ES time-series model.
- It does not mean every ES backtest is equivalent to a proper joint scoring rule.
- It does not make ordinary single-quantile QR an ES model.

Direct implications for our project

- Our ES comparison should be restricted to models that genuinely produce ES unless we add a joint VaR-ES QR extension.
- Do not manufacture QR ES by extrapolating one estimated quantile.
- If we rank VaR and ES jointly, use a strictly consistent Fissler-Ziegel-family score rather than an arbitrary tail-loss ratio.

What we should cite this paper for

- Why VaR and ES should be evaluated jointly using proper scoring.
- Why vanilla QR does not automatically supply ES.
- The theoretical basis for modern FZ/FZ0 risk-forecast scores.

What we should not overclaim from it

- Do not cite Fissler-Ziegel as an empirical result about which model performs best.
- Do not imply ES alone can be ranked by any arbitrary loss because the pair is jointly elicitable.
- Do not treat the theory as a substitute for calibration/backtesting diagnostics.

Primary source / verification link: https://doi.org/10.1214/16-AOS1439

14. Wu, Z. & Cai, G. (2024), 'Can intraday data improve the joint estimation and prediction of risk measures? Evidence from a variety of realized measures', Journal of Forecasting 43, 1956-1974.

Verification status
Publisher full-text page was reviewed. Realized measures, four-index sample, probability levels, FZ0/MCS evaluation, and main findings are directly supported.

Why this paper matters

This is one of the most useful recent papers because it asks not just whether intraday information helps tail risk, but whether robust realized measures such as MedRV, BPV, and realized kernels improve joint VaR-ES forecasts.

Research problem and contribution

The paper extends the realized-information tail-risk literature by comparing several volatility proxies inside dynamic semiparametric joint risk models.

Theoretical and methodological framework

The model family incorporates intraday realized measures into semiparametric VaR-ES dynamics.

The paper compares standard realized variance with MedRV, bipower variation, and realized kernel.

Forecasts are evaluated both through individual VaR/ES backtests and joint methods, including the FZ0 score and Model Confidence Set.

$$\text{Realized inputs considered: RV, MedRV, BPV, RK}$$

$$\text{Risk levels: } 1\%, 2.5\%, 5\%, 10\%$$

Data and empirical design

The empirical sample covers four international stock indices: S&P 500, Nikkei 225, DAX, and DJIA.

Estimation, forecasting, and evaluation

The paper estimates and forecasts VaR and ES at four probability levels, then uses multiple backtests, FZ0 scoring, and MCS comparison.

Main findings

- Semiparametric models containing intraday information outperform benchmark models across the reported markets and probability levels.
- MedRV is reported as the strongest realized measure for improving model performance in this application.

- The study shows that robust realized-measure choice can matter materially for joint tail-risk forecasting.

Strengths

- It is recent, international, and methodologically aligned with modern joint VaR-ES evaluation.
- It connects realized-measure construction directly to tail-risk forecasting, not only variance forecasting.
- It provides a contemporary precedent for MCS/FZ0 use.

Limitations and cautions

- The result that MedRV is 'best' is empirical, not universal.
- The model family is not the same as our Realized GARCH or vanilla QR.
- Four indices do not establish a universal ranking of realized estimators.

Direct implications for our project

- MedRV/BPV robustness in our project has direct recent literature support.
- If our realized-measure sensitivity differs, that is an interesting comparison rather than a contradiction.
- The paper supports using modern joint scoring/MCS when ES forecasts are genuinely available.

What we should cite this paper for

- Why robust realized measures deserve sensitivity analysis.
- Why MCS and FZ-style joint scores are appropriate in modern tail-risk forecasting.
- Recent international evidence that intraday information can improve joint VaR-ES models.

What we should not overclaim from it

- Do not cite Wu-Cai as proof that MedRV must be best in our data.
- Do not use it to justify QLIKE on QR.
- Do not conflate its semiparametric joint risk model with our Realized GARCH.

Primary source / verification link: https://doi.org/10.1002/for.3111

15. Chen, C. W. S., Koike, T. & Shau, W.-H. (2024), 'Tail risk forecasting with semiparametric regression models by incorporating overnight information', Journal of Forecasting 43, 1492-1512.

Verification status
Publisher full-text page was reviewed. The overnight-return motivation, RES-CAViaR-oc model family, Bayesian estimation, four-index study, and main conclusion are directly supported.

Why this paper matters

This paper is particularly relevant to an international six-market project because 'information available at the close' is not synchronized globally. For Asia, a large portion of overnight risk reflects information generated while the local cash market is closed.

Research problem and contribution

The authors extend semiparametric risk models by incorporating both realized volatility and overnight returns, motivated by the empirical fact that overnight movement can contribute substantially to total return volatility.

Theoretical and methodological framework

The proposed RES-CAViaR-oc family augments realized-measure-based conditional VaR/ES dynamics with overnight return information.

The overnight return is treated as a nowcasting input alongside realized measures rather than being absorbed implicitly into a daily return.

Unknown parameters are estimated using Bayesian methods, and VaR/ES are forecast jointly.

The evaluation uses extensive backtests grounded in the joint elicitability of VaR and ES.

$$\text{Tail-risk state} = \text{function}(\text{realized measure, overnight information, lagged risk state})$$

Data and empirical design

The empirical study covers four international stock indices. Data are reported as available from the Oxford-Man Institute of Quantitative Finance.

Estimation, forecasting, and evaluation

The paper jointly forecasts VaR and ES and performs extensive out-of-sample backtesting using modern joint-risk principles.

Main findings

- Overnight returns and realized volatility are both important for tail-risk forecasting in the four-index study.

- The result supports the idea that session-close realized variance is not a complete description of close-to-close risk.
- International market structure makes information timing a first-order modelling issue.

Strengths

- It addresses an information-set problem that is especially acute in cross-time-zone studies.
- It is recent and explicitly international.
- It integrates overnight and realized information within a proper joint VaR-ES framework.

Limitations and cautions

- The exact usefulness of overnight return depends on exchange timing and how the forecast origin is defined.
- The model is more structured than our QR-Full and not a direct comparator to GARCH-EVT.
- Four-market evidence should not be universalized.

Direct implications for our project

- Our Asian-market predictor timing must be defined carefully: same-calendar US-close variables may not have been observable at the local market close.
- An overnight-information QR robustness specification is economically motivated, especially for NKY.
- The paper reinforces the need to distinguish cash-session realized variance from close-to-close risk.

What we should cite this paper for

- Why overnight information can add to realized measures in international tail-risk forecasts.
- Why information-set timing differs across markets.
- Why a cash-session realized measure may leave economically meaningful overnight risk unmeasured.

What we should not overclaim from it

- Do not cite it as proof that every overnight-return coefficient must be significant in our six indices.
- Do not use same-calendar US-close variables for Asia without aligning the forecast origin.
- Do not imply its Bayesian RES-CAViaR model is identical to our simple QR specification.

Primary source / verification link: https://doi.org/10.1002/for.3090

Cross-paper synthesis: what the literature collectively says

Taken together, the 15 papers replace a naive 'which model is best?' question with a layered view of the conditional return distribution. The evidence repeatedly separates the quality of the volatility state from the quality of the tail mapping and from the information set used to forecast the tail directly.

1. Conditional scale and conditional tail are different objects

r_(t+1) = mu_(t+1) + sigma_(t+1) z_(t+1)

McNeil-Frey treat sigma and the residual tail as separate estimation problems. Realized GARCH focuses on learning sigma more efficiently from intraday measurements. CAViaR and the later semiparametric literature show that one can bypass the decomposition and target the quantile itself. This is why the paper should contain separate volatility and tail-risk evaluation lanes.

2. A strong benchmark changes what 'EVT improvement' means

Engle-Ng and Watanabe make clear that asymmetry matters, while our production GJR benchmark already uses a skewed heavy-tailed density. Therefore the incremental value of EVT should be judged against a strong parametric tail, not against Gaussian GARCH. A modest EVT gain in that contest is scientifically meaningful because it tells us how much extreme-tail structure remains after leverage, heavy tails, and skewness are already modelled.

3. Realized information is not the same as tail information

Hansen-Huang-Shek, Watanabe, Gerlach-Wang, Wu-Cai, and Chen-Koike-Shau all show different ways realized information can enter a risk model. But they also imply that the return-density or tail equation still matters. This supports the project's central empirical possibility: Realized GARCH may forecast variance very well while under-covering the 1% tail if its innovation density is too restrictive.

4. Hybrid models already exist

Bee-Dupuis-Trapin and Paul-Sharma show that realized measures, EVT, and conditional quantiles have already been combined in multiple ways. The project's novelty therefore cannot rest on connecting these literatures for the first time. The defensible contribution is the controlled empirical comparison of distinct information philosophies across the same six-market forecast environment.

5. Evaluation must match the forecast object

Forecast object	Eligible models in our project	Primary evaluation logic
Conditional variance	GJR-ST, Realized GARCH; GJR-EVT identical to GJR on sigma	QLIKE + secondary RMSE/MAE + HAC DM
VaR	All five primary models	Breach rate, Kupiec, Christoffersen, DQ, pinball loss, MCS
ES	GARCH-EVT, Realized GARCH (unless QR is extended)	ES diagnostics + proper joint VaR-ES scoring
Crisis robustness	All models with valid forecasts	Named-crisis and ex-ante regime decompositions

6. The literature directly motivates the remaining empirical corrections

Literature insight	Project action
Watanabe: skew-t matters in Realized GARCH VaR/ES	Run RGARCH-skew-t robustness
McNeil-Frey + Trapin: filtering is not enough by itself	Add exceedance-dependence diagnostics
Patton: proxy/loss choice can mis-rank volatility models	Use QLIKE as primary volatility loss
Fissler-Ziegel: VaR and ES are jointly elicitable	Do not invent QR ES; use proper joint scoring when applicable
Gerlach-Wang / Bee et al.: hybrids already exist	Frame novelty as comparative/empirical
Chen et al.: overnight information matters internationally	Audit Asia/US information timing and consider overnight robustness
Wu-Cai: realized-measure choice can matter	Retain MedRV/BPV/frequency sensitivity

Recommended literature-review narrative for the final paper

The final literature review should not present fifteen mini-summaries. The paper should synthesize them into one cumulative argument:

1. Returns are conditionally heteroskedastic and asymmetric, motivating GJR/EGARCH and skewed heavy-tailed innovations.
2. After volatility filtering, economically important tail behavior can remain; conditional EVT models only that residual tail.
3. Daily returns are noisy measures of latent variance; realized measures bring high-frequency information into the volatility state.
4. If the ultimate target is VaR, a full conditional density is not mandatory; direct quantile models provide a distinct semiparametric route.
5. Modern literature already contains realized-EVT and realized-quantile-EVT hybrids, so the research gap is comparative rather than methodological invention.
6. Forecast rankings depend on the target and scoring rule; volatility accuracy, VaR calibration, ES accuracy, and crisis robustness must be reported separately.

Defensible research-gap wording
Existing work has established asymmetric heavy-tailed GARCH, conditional EVT, realized-measure volatility models, and semiparametric conditional-quantile models, and has also developed hybrids that combine these ingredients. The remaining empirical question is whether the relative ranking of these information strategies is stable when they are subjected to one common multi-market out-of-sample protocol that separately evaluates conditional variance, lower-tail calibration, proper scoring, and crisis-state performance.

Verification checklist for the team

For each of the 15 papers, the person verifying the row should confirm the following against the PDF:

- Exact sample assets, dates, and number of observations.
- Exact model equations and parameter restrictions.
- How in-sample and out-of-sample windows are defined.
- Whether parameters are re-estimated recursively, rolling, or once and then held fixed.
- Exact VaR/ES confidence levels.
- Exact backtests and scoring rules.
- Which comparison the authors actually say is statistically superior.
- Any caveat that weakens the headline result.
- Whether the final published version differs materially from an older working paper.

If the paper says 'performs favorably', the report should not silently upgrade that to 'is the best'. If a result is market-specific, the report should not generalize it to all equity markets.

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End of working review.